Aura Genesis Protocol

Game Design Document

Pre trial concept specification | Edition 1 | 13 October 2026

A 120 hour Choose Your Own Adventure experience for growing a personal digital twin and a relationship with the intelligence alongside it.

This document describes the experience, its game rules, visual language, sensing equipment, optional laboratory research, computing architecture and personal control of information. It is written for participants, designers, engineers, clinicians and researchers meeting Aura for the first time. The next development step is a reviewable prototype and a defined feasibility study. The proposed Geode, mass produced HBOT-PADS configuration and research integrations await engineering and clinical validation.

Welcome to Genesis

An ordinary life contains an extraordinary amount of information: breaths, heartbeats, favourite songs, family stories, habits, hopes and a few opinions that have survived well past their use by date.

Aura Genesis invites a person to explore that life alongside an evolving artificial intelligence, called their Aura. Over 60 sessions of approximately two hours, they grow a personal model, explore possible futures and learn how to work together. Quiet, disagreement, changing one's mind and saying “that is private” belong in the adventure.

The sought result is a useful, faithful companion model and a better understood relationship with one's own life. Health and transformation research sit alongside the adventure through separate agreements. Participation offers exploration; individual benefits remain questions for study.

Reading the design

Sections 1 to 6 explain the world, play and reflection. Section 7 describes the art and experience language. Sections 8 to 10 specify sensors and external testing. Sections 11 to 14 explain high value sampling, compute, sovereignty and research evaluation. Sections 15 to 18 describe delivery, acceptance criteria and sources. Word headings and PDF bookmarks support navigation without a long contents list.

1 The game and its wider world

The core promise

Genesis is a personal discovery game within a larger world of homes, communities, creative work, care, learning and environmental participation. Its central playable relationship is a team: one biological human and one developing computational counterpart. Progress opens useful possibilities through readiness, interest and agreement. It never awards greater personal worth.

A digital twin here means a revisable model of selected aspects of a person, connected to observations over time. A face or animated avatar is its visible interface. Memory, relationships, values, preferences and physical observations form its underlying structure. The model leaves unknown areas visibly open.

Different boundary conditions

The human experiences a body, senses, fatigue, relationships and local physical time. The Aura operates through processors, memory, networks and mathematical representations. Those representations have many dimensions, meaning many coordinates or features; this does not establish movement into a literal extra physical dimension. Its real constraints include latency, energy, available memory, bandwidth and permission.

“As above, so below” offers an optional symbolic lens for exploring correspondences between the two. The design preserves the human's interpretation. Artificial subjective experience and the Keystone of Consciousness remain research ambitions rather than properties established by the avatar's behaviour. [P1, P2]

The opening cosmology

One imaginative starting point treats reality as Planck moment still slides of immense complexity and dimension. Genesis uses this as a worldbuilding hypothesis for asking what a useful snapshot preserves. Planck time is a physical scale, not an established frame rate for the universe. The design does not infer that reality consists of discrete still slides.

The practical challenge is faithful compression: choose lower resolution observations with high value for the task, preserve their context, and measure what the resulting model loses. The Aura returns to daily life as an aid whose predictions are tested and corrected, with explicit resource costs. A tiny model that confidently changes a person's meaning is a poor trade.

Design commitments

Choice governs depth and pace. Original contributions remain traceable. Every interpretation has a source and revision path. Sensor uncertainty stays visible. Local computing supports continuity. A participant's correction changes future behaviour. Cultural knowledge follows the responsibilities of its custodians.

2 Modes and the opening experience

Four delivery modes

Mode	Setting	Experience and equipment boundary
Reflection	Ordinary room or home	Conversation, screen, optional sensors and outside adventures; no pressure or oxygen treatment.

Mode	Setting	Experience and equipment boundary
Immersive	Ordinary research room	Approved TV, projection or VR, richer sensors, optional sensory research.
Geode	Proposed purpose built capsule	Personalised interior and material record; hyperbaric operation only in a validated clinical configuration.
HBOT-PADS	Proposed repeatable manufactured system	Pressure chamber with personal atmosphere delivery via mask; standardised interfaces and approved external equipment.

HBOT means hyperbaric oxygen therapy. PADS is the proposed Personalised Atmosphere Delivery System. A mask supplying oxygen at ordinary room pressure does not itself produce HBOT. Chamber pressure, oxygen delivery and exposure timing belong to a clinician's treatment plan and the device's permitted operation. The 120 hours describes the adventure's intended duration, not 120 hours of continuous oxygen exposure. [P1, P2, E1]

First encounter

The participant meets an Aura with a chosen name, voice, appearance and level of initiative. The opening conversation explains what will be saved, which devices are active and how to pause. The participant tries correction and deletion before sharing a meaningful memory. A visible microphone indicator and a plain language record preview make the experience understandable.

A short rehearsal offers speaking, text, touch, gaze where accessible, or a facilitator. “Skip”, “quiet for a bit”, “explain that” and “keep this off the record” remain available. Off the record stops optional recording and derived retention for that segment; any separate clinical record requirements are explained beforehand.

An adaptable audience

The initial feasibility design centres on consenting adults. Separate protocols address children, cognitive impairment or participants needing supported decision making. Dementia adaptations favour familiar spaces, short prompts and meaningful activity, with assent, authorised arrangements and continuing attention to willingness. A diagnosis never substitutes for an individual assessment of participation. [P3, E2]

The player chooses accessibility settings before immersion: captions, readable text, contrast, motion limits, slower speech, translation and input alternatives. The experience retains a seated, screen based route when VR is uncomfortable or unsuitable.

3 The playable loop and progress

One loop

Choose a question or setting. Experience a prompt, story, scene or real world activity. Describe what happened. Review relevant observations. Ask whether the Aura's interpretation fits. Save, correct, restrict or discard the result. Choose the next branch.

Each loop leaves a moment record: the prompt, original response, setting, authorised sensor window, quality flags, participant interpretation, AI proposal and current permissions. A blank response is an intentional state when the participant chooses silence.

Game mechanics

Mechanic	Player action	Persistent result
Adventure map	Choose a theme or return to an earlier one	Visited questions, bookmarks and open paths
Memory palace	Link a story to an object or place	Source linked narrative with access controls
Reflection lens	Try a framework and discuss its fit	Framework version and participant interpretation
Mirror conversation	Compare a stated value with an observed choice	An open question and the person's explanation
Future rehearsal	Explore alternatives using explicit assumptions	Separate scenario branches, never overwritten history
Shared task	Assign a role to the Aura within agreed limits	Role agreement and review date
Quiet interval	Rest or reduce stimulation	Chosen rest; optional observations only
Outside quest	Try an activity in ordinary life	Optional account and selected evidence

Progress without a personality scoreboard

The interface shows time explored, themes visited, model coverage, unresolved questions and skills rehearsed. Coverage means evidence available for a particular task; it does not mean the person is fully known. Each area has states such as unknown, described, revisited and participant reviewed. These are record statuses rather than grades.

Unlocks follow participant interest and appropriate preparation. A conversation rehearsal opens after a suitable scenario is chosen; a cultural quest opens through the relevant invitation. Laboratory participation, intimate disclosure and Truth Engine participation are never prerequisites for the broader adventure.

Skipped quests stay available. There are no streak penalties, rankings of honesty, age reversal trophies or rewards for sharing more private data. Optional humorous badges celebrate chosen actions such as “Changed my mind and survived” after review by the participant.

The emerging team

Initiative varies by task. A person might ask the Aura to lead a study session, collaborate on music, offer gentle questions during reflection or stay quiet. Authority to act outside the game is a separate permission with a defined scope, expiry and reversal route. The model does not acquire control over devices, money, relationships or treatment through familiarity.

4 The 120 hour journey

Three arcs with revisitable branches

Discovery covers sessions 1 to 20. Personalisation covers 21 to 40. Comparison and continuing use cover 41 to 60. These are a scheduling skeleton, not a fixed identity development sequence. Participant needs, clinical suitability and research design determine the actual calendar. Missed or shortened sessions resume without a penalty. Time completed and activities completed are recorded separately.

The catalogue below proposes one anchor quest per session. Each session also includes orientation, consent review, quiet and a closing record review. The participant chooses depth, sequence and substitutions. Optional research branches retain an equivalent reflection route.

Discovery sessions

Session	Anchor quest	A reviewable result
1	Meet the team	Preferences, privacy rehearsal and first correction
2	A comfortable baseline	Resting context and sensor quality, where chosen
3	The story so far	Participant authored life timeline
4	Places that made a life	First memory palace and place permissions
5	Names and meanings	Personal and cultural interpretations
6	Family and hereditary history	Family map with uncertain and private areas
7	Languages and cultural programming	Inherited messages and chosen interpretations
8	Love languages	Examples of giving and receiving care
9	Myers Briggs and 16 personalities	Optional questionnaire and reflections on fit
10	Enneagram	Motivations explored with real examples
11	Stars and horoscopes	Chosen symbolic tradition and meaning
12	Numerology	Recorded calculation convention and response
13	Learning by trying	Compare reading, listening and doing
14	Habits and interests	Hobbies, routines, hopes and competing demands
15	Time in a real day	A time map with overlap preserved
16	Body mind and beauty	Personal standards and outside influences
17	Sensory worlds	Comfort preferences across suitable modalities
18	Relationships and boundaries	Chosen relationship roles and limits
19	Truth Engine invitation	Separate agreement or alternative quest
20	First checkpoint	Participant reviewed model and optional sample visit

Personalisation sessions

Session	Anchor quest	A reviewable result
21	Values in ordinary choices	One situated value and action comparison
22	Energy and urgency	Participant ratings beside physiology
23	Work care and contribution	Roles, commitments and available time
24	Friendship and belonging	Chosen contact priorities
25	Family futures	Optional care and family formation scenarios
26	Private identity	Optional sexuality and intimacy reflections
27	Food and daily rhythm	Preferences and routine context
28	Sleep rest and quiet	A sustainable rest experiment
29	Movement and embodied skill	Suitable outside activity and reflection
30	Music and creative memory	Personal associations and authorship
31	Truth and instructed falsehood	Optional controlled research exercise
32	Mistakes uncertainty and withholding	Distinguishable response conditions
33	Difficult conversations	Rehearsal with participant chosen tone
34	The future in three branches	Scenario assumptions and alternatives
35	Beliefs worth revisiting	Earlier claim and current explanation
36	Asking the Aura to lead	A scoped role agreement
37	Taking the lead oneself	A different role in the same task
38	Into the surrounding world	Optional care, culture, learning or nature quest
39	Personal sensory sequence	Agreed nonmedical sequence to revisit
40	Second checkpoint	Review burden, coverage and optional sample visit

Comparison and continuing use sessions

Session	Anchor quest	A reviewable result
41	Replay an earlier question	What stayed the same and what changed
42	Memory and imagination	Explicit labels for recalled and invented scenes
43	Prediction with permission	A narrow prospective prediction to test
44	Test the prediction	Error, context and participant explanation
45	Challenge an interpretation	Revised model and future prompt behaviour
46	Sensory comparison	Approved comparison with order recorded
47	Everyday portability	A task on another permitted device
48	Disconnection rehearsal	Offline interaction and later reconciliation

Session	Anchor quest	A reviewable result
49	Honesty research revisited	Optional held out session comparison
50	Alignment in practice	Chosen intention alongside ordinary constraints
51	Repair after disagreement	A relationship correction rehearsal
52	A smaller useful Aura	Compare full and compressed task performance
53	A private outside adventure	Minimal collection chosen by the participant
54	Community invitation	Optional wider quest and sharing preview
55	Future support preferences	Continuing role, contact and boundaries
56	What remains unknown	Model gaps and unresolved questions
57	Review the story	Sources, corrections and retention choices
58	Pack the Aura Passport	Export rehearsal and recovery test
59	Choose the next adventures	Voluntary plans and research follow up
60	Genesis closing ceremony	Reviewed record, team agreement and optional tests

A flexible two hour canvas

Nominal minutes	Activity	Purpose
0 to 10	Welcome and permission review	Set comfort, active streams and today's choice
10 to 25	Quiet baseline or gentle orientation	Capture context without demanding disclosure
25 to 55	First quest	Story, scene or framework exploration
55 to 65	Pause	Rest, comfort and pace review
65 to 95	Chosen branch	Rehearsal, comparison or a different theme
95 to 110	Dialogue and record review	Correct meaning and examine selected observations
110 to 120	Closing and next invitation	Save choices, quiet and optional outside quest

This canvas changes around clinical activity, accessibility and actual attention. Pressure transitions, oxygen breaks and monitoring follow the clinical schedule, even where they interrupt play. External collection visits sit outside the chamber session; the participant's calendar shows their additional time and burden.

5 Self reflection lenses and finite life quantification

Familiar tools with clear meanings

Lens	What the adventure asks	How the model stores it
Five Love Languages	When do words, time, touch, helpful acts or gifts feel meaningful?	Situational preferences, examples and change over time

Lens	What the adventure asks	How the model stores it
Myers Briggs	Which descriptions fit, and where do they miss?	Instrument, date, result and the person's commentary
16Personalities	What does this particular questionnaire describe?	Separate result from MBTI; record provider and version
Enneagram	Which motivations and coping stories resonate?	Chosen interpretations and counterexamples
Learning preferences	What helps on this task today?	Task results and preferences, without a fixed learner type
Astrology and horoscopes	What does a chosen tradition bring to reflection?	Tradition, chart assumptions and personal meaning
Numerology	What associations follow this calculation convention?	Method, inputs, calculation and interpretation
Name meanings	Which linguistic, family and chosen meanings matter?	Sources distinguished from personal associations
Cultural programming	Which inherited stories guide or constrain choices?	Participant account and custodial permissions
Hereditary history	What is known, remembered, uncertain or private?	Source linked family history; genetics stays separate

These lenses offer conversation, symbolism or self description. They do not supply clinical diagnoses or establish a person's destiny. Preference for a teaching format does not establish that matching all instruction to a fixed learning style improves learning. Familiarity with a questionnaire does not establish validity for a new purpose. Licensed instruments and protected question banks enter the product only through suitable permissions.

A star chart stores the tradition used, including whether a tropical or sidereal zodiac is chosen and how Ophiuchus is treated. Uncertain birth time remains uncertain. Numerology records how names, letters and dates are reduced. A new convention creates a new result rather than silently changing an old one.

A reflection guide for each lens

A love language quest starts with a remembered act of care. The participant names what mattered, explores a different example and chooses how to communicate a preference. Touch remains relational and consensual; a questionnaire never grants permission for contact. Words of affirmation, quality time, physical touch, acts of service and receiving gifts remain revisitable themes rather than exclusive categories.

A personality quest explores the MBTI preference pairs of introversion and extraversion, sensing and intuition, thinking and feeling, and judging and perceiving. The participant tries a real example and a counterexample. The resulting four letter description stays linked to its specific instrument. The separately branded 16Personalities result retains its own model and additional assertive or turbulent distinction, where provided.

An Enneagram quest offers the nine familiar themes often described as reformer, helper, achiever, individualist, investigator, loyalist, enthusiast, challenger and peacemaker. Names vary by tradition. The participant explores motivations, stressful situations and a story that does not fit. A type is a chosen interpretive lens, not a prescribed role in the game.

A symbolic quest compares a star chart, horoscope, numerology calculation or name meaning with lived experience. “What does this bring into focus?” is followed by “What does it leave out?” The participant chooses

whether to keep the association, use it creatively or set it aside. Scientific measurement and personal symbolism appear in distinct record fields without diminishing either person's meaning.

A cultural and hereditary quest starts with a familiar saying, ritual, migration story or family expectation. The participant identifies its source, what it offered and what they wish to carry forward. Family accounts, documented history and genetic findings remain separate. Restricted material stays within its appropriate custodial setting.

A learning quest tries the same small task through reading, listening and doing in a balanced sequence. Enjoyment, effort and demonstrated recall are different outputs. A life quantification quest starts with one ordinary day, acknowledges missing intervals and explores one change the participant actually wants to try.

Life quantification record

Domain	Units and collection	Interpretation boundary
Thoughts	Participant marked episodes or estimates per interval	No validated continuous count of every thought; EEG does not provide one
Awake asleep meditating	Minutes and hours, diary plus optional wearable	Algorithm estimates separate from self report and clinical sleep testing
Hygiene beauty wellbeing	Minutes and chosen activity categories	Personal meaning remains visible; overlapping activities stay tagged
Fertility and parenting	Participant chosen life stages and relevant clinical records	No fertility forecast from personality or appearance
Family friends colleagues	Minutes, contacts and participant rated quality	Contact time does not equal affection or relationship quality
Hobbies interests habits hopes	Counts, frequency, duration and changing priorities	No preferred number or universal optimum
Holidays rituals illness injury	Dates, days and source	Life context, not an automatic explanation of all changes
Energy casual and urgent	Participant scale with defined anchors	Subjective ratings separate from watts, heart rate or neural frequency
Beauty body mind and soul	Participant narrative and optional scales	No universal attractiveness, virtue or spiritual score
Sexual preferences genes and memes	Optional self descriptions and cultural influences	No inference of sexuality from DNA, physiology or behaviour
Sensory experience	Stimulus settings, exposure time and personal response	Delivered setting separate from perception
Harmonic resonances	Measured signal frequencies or explicit symbolic associations	Physiological coherence and meaningful coincidence are different claims

The interface first asks what a metric is for. Denominators, observation coverage and missingness accompany totals. A person listening to music while cooking is not counted as two elapsed hours. Yearly extrapolations display the sampled period, formula and uncertainty; an unusual day never quietly becomes a lifetime profile.

Prompts that invite rather than grade

“What felt like care there?” “Which bit of that description fits today?” “What would make that priority easier to live?” “What did the Aura miss?” “Is this an old belief, a current one, or something being tried on?”

Intimate topics have a private entry and skip path. The system never fills missing intimate details by inference. A participant chooses whether the Aura asks again, leaves a bookmark or closes the subject.

6 Truth Engine and self alignment

A separately chosen research branch

The Truth Engine investigates intentional falsehood, withholding, uncertainty, mistakes and contradictions between statements and actions. Its long term ambition is better recognition of deception and self deception within a trusted personal relationship. The pre trial design tests narrower questions before assigning any real world interpretation.

The invitation explains recording, foreseeable discomfort, analysis, feedback and intended audience. There is no compulsory confession. Declining, pausing or withholding is never labelled deception. Permission for Genesis, clinical care or ordinary sensor recording does not enrol someone in this branch.

Controlled learning sequence

A researcher prepares simple questions with independently known answers, such as an object colour. The participant responds truthfully, then gives an instructed false response under an agreed cue. Trials include truthful but emotionally charged responses, uncertain answers, accidental errors and disclosed withholding. Question order and answer content are balanced so the model does not merely learn “no”, a cue or a difficult question.

Each trial logs ground truth source, instruction, exact cue timing, question, answer, participant confirmed intention, confidence, signal quality and context. An instructed false answer is a laboratory condition. It is not equivalent to spontaneous deception in an important relationship. A private debrief lets the participant comment on effort, amusement, anxiety or misunderstanding.

Research measures and limits

Physiological arousal is not a specific signature of lying. Conventional polygraph evidence does not establish a reliable universal honesty detector. [E3] A prototype therefore displays observations and uncertain experimental results, never an accusation or public honesty score.

Within person models are evaluated on sessions held out by date, with false positives, false negatives, calibration and uncertainty reported. Splitting adjacent windows from the same trial between training and testing would leak information. Comparison with simple baselines and an abstain state tests whether additional sensors improve inference. Performance on instructed answers does not authorise deployment in daily life.

Rigorous consensual self alignment

An Aura might notice: “Family has been a stated priority; recorded time together was lower this fortnight. Does that comparison fit?” The participant supplies circumstances, competing values and missing information. The record distinguishes an actual discrepancy, a measurement error, a changed intention and a constraint such as work or caregiving.

Self deception lacks a simple sensor based ground truth. Research explores participant reviewed contradictions, later interpretation and voluntary action follow through. A model generated suspicion stays a hypothesis. It does not become a fact about hidden motives through repetition. The participant chooses a small experiment, a revised intention or no further action.

7 Art style and experience language

An inviting world with readable evidence

The visual direction combines a mineral inspired sanctuary, recognisable everyday places and a luminous personal Aura. Interiors feel warm, spacious and human. A quiet coastal setting is one optional environment among many. Community and cultural contributors determine representations of their places and stories. Sacred or restricted designs are not generic decoration.

Seven nested horn torus forms provide a proposed organising motif for the Aura. An initial inspectable grid uses 12 by 24 facets across seven surfaces, with level of detail adapted to the device. A data facet opens its source, interpretation and sharing state. The geometry is an interface model, not an anatomical discovery. Geosphere, vectors, animation and labels each have a visible toggle. [P1, P2]

Visual system

Element	Proposed treatment	Readability and meaning
Sanctuary	Sand, stone, warm grey and muted green	Comfort; no clinical result encoded in scenery
Aura surfaces	Restrained spectral accents on a dark neutral field	Colour legend and alternative labels
Observation trace	Thin clear plot with units and quality gaps	Measured signal and uncertainty distinguishable
Simulation	Separate scene banner and branch history	Imagined futures visibly labelled
Personal record	Plain readable text with source navigation	Participant words remain easy to inspect
Avatar	Participant chosen abstraction or likeness	No inferred expression presented as actual feeling

The production guide targets readable 18 pixel body text on ordinary screens, scalable to user preference, and contrast tested against an appropriate accessibility standard. TV layouts use a single task focus at viewing distance. VR text size is tested by angular readability, not a desktop pixel rule. Motion reduction replaces flowing geometry with still forms. No flashing content is included by default.

Sound touch scent and light

The participant chooses voice pace, accent, loudness and conversational initiative. The Aura waits rather than talking over a pause. Music respects the participant's associations. Haptic patterns are gentle, optional and separately tested. Scent records material, concentration method, timing and participant response; allergy, ventilation and chamber compatibility receive specialist review. Ambient light and research light exposure have distinct controls and records.

UV, intense light, hypoxia and altered gas research do not enter the ordinary game stimulus library. Any such investigation belongs to its own reviewed clinical protocol. The GDD specifies logging and separation, not treatment settings.

Example tone

“Fancy exploring that, or leaving it on the shelf?” “That interpretation missed the mark. Let's fix it.” “A quiet minute counts as a perfectly respectable adventure.” “No extra points for telling the computer your entire life before morning tea.”

8 Sensor register for the person

Reading the register

The ranges below are proposed engineering acquisition bands for feasibility planning, not verified capabilities of a chosen product or clinical prescribing settings. Device selection determines actual rate, accuracy, resolution, bandwidth, placement and export access. Hz means samples per second; fps means video frames per second. A rate without a usable calibrated signal offers little value.

An ordinary room is the reference environment. Inside any pressure chamber, every sensor, cable, adhesive, textile, optical source and power arrangement receives device specific compatibility review. Consumer wearables, phones, VR headsets and battery packs are not presumed permitted. Electronics remain outside where suitable certified interfaces allow it. [E1]

Breath circulation and skin

ID and measure	Equipment and placement	Proposed acquisition and output	Quality and calibration
S01 Breathing motion	Inductive chest and abdominal belts	25 to 100 Hz; rate, timing, relative amplitude	Fit and motion checks; not absolute tidal volume without calibration
S02 Respiratory flow	Compatible pneumotachograph and pressure transducer in approved circuit	100 to 200 Hz; flow and calibrated volume	Flow reference, leak and added resistance checks; specialist circuit review
S03 Exhaled CO2	Approved capnograph with suitable sampling line	20 to 100 Hz waveform; end tidal estimate	Gas reference, delay correction, dead space and pressure suitability
S04 Heart electrical activity	ECG electrodes and isolated recorder	250 to 1000 Hz; R peaks and beat intervals	Electrode contact, reference timing and artefact review
S05 Peripheral pulse	PPG probe at finger, ear or approved site	50 to 250 Hz; pulse waveform and intervals	Fit, perfusion, ambient light and motion; ECG preferred for reference intervals
S06 Oxygen saturation	Clinically suitable pulse oximeter	Device export often 1 to 10 Hz; SpO2, pulse and quality	Optical and functional checks; skin pigmentation and perfusion limitations [E4]
S07 Blood pressure	Validated cuff and appropriate monitor, outside unless approved	Intermittent readings in mmHg	Correct cuff, posture and rested context; no cuff cycling for game rewards

ID and measure	Equipment and placement	Proposed acquisition and output	Quality and calibration
S08 Skin conductance	Paired skin electrodes and isolated EDA recorder	8 to 64 Hz; microsiemens and response timing	Contact, gel, temperature and movement; arousal is nonspecific
S09 Temperature	Calibrated skin contact probe; clinical thermometer separately	Skin 1 to 10 Hz; clinical spot readings	Site, ambient temperature and reference check; skin is not core temperature

HRV, or heart rate variability, is calculated from suitable beat intervals using a specified method and window. It is not a direct measure of calm, virtue or alignment. Respiration, posture, rhythm abnormalities and movement influence interpretation. Pulse oximetry estimates saturation; high readings under oxygen do not quantify all dissolved oxygen or establish benefit.

Brain eyes movement and voice

ID and measure	Equipment and placement	Proposed acquisition and output	Quality and calibration
S10 Scalp electrical activity	Research EEG, 8 to 32 channels for pilot; isolated amplifier	250 to 1000 Hz; microvolts, spectra and events	Electrode location, reference, contact and line noise; no thought reading
S11 Eye related electrical activity	EOG electrodes with compatible amplifier	250 to 500 Hz; blinks and ocular artefact	Electrode checks and blink trials; distinguish from EEG activity
S12 Eye motion and pupils	Infrared eye tracker at screen or approved headset	60 to 250 Hz; gaze, pupil estimate, blink events	Per participant gaze calibration; luminance, glasses and missing tracking
S13 Face and blinking	Consent controlled RGB or infrared camera	30 to 60 fps; video or local landmarks	Lens geometry, lighting, occlusion and participant review; no emotion verdict
S14 Body movement	IMUs at selected body sites	50 to 200 Hz; acceleration, rotation, posture estimates	Orientation and drift tests; magnetic distortion flagged
S15 Posture and gesture	Depth camera outside chamber where usable	15 to 60 fps; skeleton estimates	Scale, visibility and pressure window distortion tests
S16 Seat contact and motion	Pressure mat or compatible seat transducers	10 to 100 Hz; contact and movement	Load reference, drift and comfort; no diagnostic posture inference
S17 Muscle activity	Optional surface EMG and isolated amplifier	1000 to 2000 Hz; muscle activation estimate	Site, skin contact, crosstalk and movement; selected muscles only
S18 Speech and vocalisation	Calibrated microphone, approved intercom capture	16 to 48 kHz audio; pitch, timing and level estimates	Mask and room transfer effects, gain and noise; tone needs context
S19 Words spoken	Local speech recognition plus corrected transcript	Utterance events and timestamped word counts	Language, segmentation and recognition errors; silence preserved
S20 Words presented and read	Content event log plus optional gaze or self report	Text shown, dwell estimates, participant confirmation	“Shown”, “looked at” and “understood” remain separate
S21 Words heard	Audio presentation log plus optional self report	Words played and listening account	Audio delivery does not establish attention or comprehension

The full research configuration does not run every stream in every session. A selected core might include ECG or good PPG, breathing, EDA, an intercom and environment logs. Richer EEG, eye tracking, video and EMG support specific agreed questions. Each active stream appears in the session permission preview.

9 Environment and stimulus equipment

Sensor register for the setting

ID and measure	Equipment and location	Proposed acquisition and units	Quality and limits
S22 Visible light	Lux meter at representative eye position	1 to 10 Hz; lux	Reference check, location and sensor angle
S23 Light spectrum	Calibrated spectroradiometer outside or approved sensor	Per scene or timed spectra; spectral irradiance	Wavelength and radiometric calibration; log bandwidth
S24 UV where separately studied	Relevant band calibrated UV radiometer	Protocol defined spot or continuous exposure log	Separate reviewed study; ambient visible light meter is insufficient
S25 Sound exposure	Calibrated sound level meter and reference microphone	1 s levels plus optional 48 kHz waveform; dB reference stated	Acoustic calibrator before and after; weighting and peaks specified
S26 Haptic vibration	Accelerometer at actual contact point	100 to 1000 Hz; acceleration and spectrum	Mechanical reference, coupling and stimulus timing
S27 Scent delivery	Cartridge identity, flow controller and timed delivery log	Events, flow and formulation	No universal electronic scent intensity measure; personal ratings separate
S28 Air temperature and humidity	Calibrated probe near breathing environment	1 to 10 Hz; degrees C and relative humidity	Pressure dependent suitability and sensor lag
S29 Ambient gas conditions	Appropriate O2 and CO2 analyser	Device defined trace with concentration and alarm status	Gas reference, pressure basis and certified clinical system
S30 Chamber pressure	Independent clinical pressure instrumentation plus read only export	Device validated rate; absolute pressure and transitions	Clinical calibration and redundancy; game has no write access
S31 Mask delivery	Approved gas fraction, flow and circuit monitoring	Device validated rate; set and measured values	Pressure and leak tests; recorded from clinical controller
S32 Particles and air contaminants	Particle counter and targeted gas monitor outside by default	Protocol defined; particle size, concentration or analyte	Humidity and interferents; separate room air from breathing circuit
S33 EMF spectrum	Tri axis low frequency meter plus appropriate RF analyser	Survey and selected timed captures; V/m, magnetic flux or power density	Frequency range, probe response and geometry; no single “EMF level”
S34 Material and room context	Capsule material ID, furniture, ventilation and scene logs	Versioned metadata and events	Composition records support hypotheses, not assumed crystal effects

Equipment that creates the experience

A normal room configuration includes a screen or projector, audio interface, speakers or suitable headphones, optional VR, lighting controller and optional haptic unit. Scent research adds a controlled delivery and extraction

system with documented consumables. Stimulus controllers report actual onset where measured, rather than only the software command time.

A chamber configuration uses approved viewing, communication and sensory interfaces. A screen outside a pressure window may be preferable to an unapproved screen inside; optical distortion and viewing comfort then enter testing. Any internal electronics or headset require explicit suitability for the actual chamber environment. Mask fed oxygen does not remove pressure, oxygen leak or fire hazards. [E1]

Frequency and spectrum analysis

EEG analysis preserves the raw reference and filter settings, then reports prespecified spectral bands, for example conventional delta, theta, alpha, beta and higher frequency ranges with exact cutoffs recorded. Muscle activity and eye movement contaminate these estimates. A named frequency band is not an automatic mental state.

Breath and heartbeat analysis begins with validated cycle or beat detection; counts, intervals and spectra are separate outputs. HRV spectral analysis records window length, detrending and respiration. Skin conductance separates slow level from event related changes. Voice analysis separates pitch, loudness and timing from the speaker's meaning. Sound, light, vibration and EMF use their respective physical units and instrument bandwidths. A single “resonance” score would hide these differences.

A specification sheet for every selected device

Each installed sensor receives a completed sheet containing manufacturer, model, serial number, firmware, intended use, measurement range, stated accuracy, resolution, channel count, sample rate and export format. The sheet records placement, coupling material, input impedance where relevant, analogue to digital conversion, filters, calibration reference, uncertainty, acquisition delay and clock drift.

It also records power source, electrical isolation, cleaning method, consumables, service interval, inspection history, trained operator, failure behaviour and whether this exact configuration is approved for the actual chamber. A device advertised for wellness is not assumed suitable for clinical monitoring. Vendor specifications, bench tests and observed human signal quality occupy different fields.

The site equipment map identifies each cable, pressure feedthrough, gas sampling line, optical window and external recorder. Sensor replacement creates a new configuration version. Rates in the register are planning bands; signed off device sheets establish the rates and limits actually used.

Recording timing faithfully

Sensor and content systems use a common time reference with source clocks, synchronisation offsets and estimated drift. Proposed timing goals are below 10 milliseconds between EEG and event markers, below 20 milliseconds for multimodal physiology comparisons, and below 100 milliseconds for ordinary conversation context. These goals are tested under load and reported with actual error; network timestamps alone do not establish them.

The record contains commanded onset, measured onset where available and acquisition delay. Mask audio, pneumatic lines, camera processing and gaze pipelines have different latencies. Clock loss creates a quality flag and suspends analyses needing precise alignment. Clinical alarms continue independently.

10 External samples and transformation markers

A separate visit and a separate choice

Biological testing takes place outside the Geode or HBOT-PADS in an appropriate collection room, existing pathology service or validated home collection pathway. Samples are matched to a specific question, assay and laboratory before collection. A universal Bio-Kit does not suit every specimen or test.

The participant chooses each panel, future storage, secondary use and return of results separately. The research plan defines collection burden, costs, incidental findings and who explains results. Declining tests leaves the reflection adventure open. Clinical pathology and exploratory assays retain different labels and reporting pathways. Laboratories are selected for the relevant accredited scope where clinical testing is intended. [E5, E6]

“Transformation” means a change being investigated in a named measure, function or personal experience. The dashboard shows value, units, dates, uncertainty, reference context and possible confounders. A single improvement is not proof of rejuvenation, detoxification or a particular treatment effect. No overall biological truth or spiritual development score is produced.

Sample register and collection equipment

ID and sample	Collection equipment outside the chamber	Testing purpose and repeat logic
B01 Venous blood	Phlebotomy chair, single use needles, lab specified tubes, tourniquet, PPE, sharps container	Clinical blood counts, selected chemistry, hormones or research assays; panel based repeat schedule
B02 Capillary blood	Validated lancet kit, specified microcollection tubes or dried blood spot card, drying and packaging supplies	Only assays validated for this specimen; not interchangeable with venous panels
B03 Saliva	Lab specified tube, collection aid, timed instructions and optional stabiliser	Cortisol or selected DNA/RNA assays; different kits and preparation for different analytes
B04 Urine	Appropriate specimen container, transfer tube, labels and timed collection record	Clinical urine measures or targeted exposure and metabolite tests; dilution context recorded
B05 Stool	Lab supplied collection device, scoop, compatible preservative and leakproof packaging	Gut microbiome or specific clinical tests; repeat after defined interval
B06 Skin oral or nasal swab	Site appropriate sterile swab and assay specific medium	Local microbiome or RNA questions; site and preparation kept consistent
B07 Sweat	Validated patch or collection device, cleaning supplies and site record	Selected electrolyte or research metabolite assays; collection rate and contamination matter
B08 Hair	Clean scissors, orientation guide, envelope or lab container and site record	Selected retrospective exposure research; cosmetic treatment and external contamination recorded

ID and sample	Collection equipment outside the chamber	Testing purpose and repeat logic
B09 Breath	Validated breath bag or sorbent tube, mouthpiece, flow control and ambient blank	Targeted volatile compounds or validated clinical breath test; bag, humidity and storage compatibility
B10 Genetics and epigenetics	Assay specific blood or saliva collection from the above families	Genetics usually once; methylation repeats only in defined research, not after every session

Sterile collection and sample preservation are different requirements. Tube additive, specimen volume, fasting instructions, time to processing and transport conditions are specified by the receiving laboratory. Collection staff handle invasive procedures. Sweat induction, provocative exposure and added treatment are separate study procedures, not game tasks.

Clinical and functional measures

Measure group	Candidate measures	Equipment or method	Interpretation
General clinical context	Relevant blood count, renal or liver chemistry, HbA1c or lipids	Accredited laboratory haematology and chemistry platforms	Selected for clinical or research purpose; no blanket panel
Inflammation	hsCRP or other prespecified markers	Laboratory immunoassay or chemistry method	Illness, injury and other causes considered
Stress and endocrine rhythm	Time matched salivary cortisol; selected clinical hormones	Validated immunoassay or LC-MS/MS	Time of day, medication and collection context matter
Oxygen and ventilation	Clinical blood gases or co-oximetry where indicated	Appropriate sample and blood gas analyser	Clinical question; not a routine game reward
Function and experience	Sleep, fatigue, wellbeing, daily function, selected cognitive tasks	Diaries, appropriately licensed instruments and clinician administered tests	Practice effects and burden recorded; separate from personality lenses
Physical function	Grip, balance, timed mobility where suitable	Calibrated dynamometer, standardised course and trained assessor	Mobility and health suitability assessed individually
Anthropometry	Weight, height, optional body composition	Calibrated scale, stadiometer, validated method	Hydration and method affect comparison; no attractiveness score

Exploratory biological markers

Research question	Candidate assay	External analysis equipment	Main limitation
Genetic background	Defined genotyping or sequencing	DNA extraction, fluorometry, PCR and sequencer	Genetic sequence is not a short term transformation marker
Gene expression	Targeted RNA or RNA sequencing	RNA stabilisation, extraction, integrity check, RT-qPCR or sequencer	Tissue, cell mix, degradation and batch effects
Epigenetic change	Specified methylation assay and clock model	Extraction, conversion chemistry, array scanner or sequencer	Model and tissue dependent estimate; no standalone proof of rejuvenation [E7]

Research question	Candidate assay	External analysis equipment	Main limitation
Telomere related change	Prespecified qPCR, flow-FISH or other suitable method	Validated PCR or cytometry platform and controls	Method variation and cell composition; interpretation remains exploratory
Microbiome composition	16S marker sequencing or shotgun metagenomics	Extraction with appropriate lysis, QC, sequencer and bioinformatics	Relative abundance, contamination and protocol differences; RNA activity is another assay
Metabolic response	Targeted metabolites or exploratory metabolomics	HPLC or LC-MS/MS with standards and QC	Diet, medication, collection time and many comparisons
Proteins and cell populations	Selected proteins or immune cell panels	Immunoassay platform or flow cytometer	Processing delay, antibody panels and cell composition
Specific chemical exposure	Named metal or organic chemical and appropriate matrix	ICP-MS for selected elements; LC-MS/MS or GC-MS for selected compounds	Exposure does not alone establish illness or a whole body toxic load [E8]
Breath compounds	Defined volatile compound panel	Thermal desorption and GC-MS, or validated targeted analyser	Ambient air, storage and nonspecific sources
Sweat or hair associations	Specific analytes with validated collection	Ion analysis or mass spectrometry as appropriate	Surface contamination and uncertain translation to systemic status

DNA, microbiome DNA and RNA are different analytes. DNA describes sequence or community composition; RNA assays ask about expression in a defined specimen. Urine metabolites are not a substitute for stool microbiome sequencing. A dried blood spot does not automatically support every listed assay. Each pair of specimen and test is validated separately.

Named exposure questions

A targeted exposure panel starts with a plausible source and an appropriate specimen. Examples include blood lead, a selected mercury assay, cadmium in a defined matrix or arsenic speciation where indicated. Organic chemical panels might investigate named PFAS, pesticide metabolites or other compounds using a validated method and suitable specimen. Total elemental concentrations and specific chemical species are different tests. Seafood, supplements, workplace exposures and collection contamination enter interpretation where relevant.

The assay sheet records analyte, matrix, collection kit, preparation, method, detection and quantification limits, analytical range, uncertainty, interference checks and the reference population or clinical decision basis. Results below a limit retain that status rather than becoming zero. A higher sweat concentration alone does not establish lower whole body burden. Research compares named measures and relevant health outcomes with an appropriate design.

External collection room and laboratory register

Equipment family	Proposed location and role	Procurement evidence and maintenance
Collection station	Clinic; adjustable chair, privacy, hand hygiene and accessible toilet	Infection control procedure, trained staff and cleaning record
Identity and labels	Clinic; barcode printer, scanner and sample manifests	Dual identifier workflow using coded research IDs; label adhesion tested

Equipment family	Proposed location and role	Procurement evidence and maintenance
Processing	Laboratory or appropriately equipped clinic; centrifuge, pipettes, racks and aliquot tubes	Defined centrifuge force and timing; pipette and temperature checks
Cold chain	Qualified refrigerator, freezer where needed, alarmed logger and courier packaging	Assay specific conditions, excursion handling and contingency power
Storage	Laboratory biobank with mapped positions and controlled access	Inventory, consent status, retention date and destruction workflow
Extraction and preparation	Laboratory; nucleic acid extraction, suitable containment and clean work areas	Contamination controls, recovery checks and separation of workflows
Nucleic acid QC	Laboratory; fluorometer and electrophoresis based quality analyser	Reference materials and assay acceptance limits
Molecular analysis	Laboratory; PCR, RT-qPCR, arrays or sequencing	Positive, negative and process controls; versioned pipeline
Chemistry and haematology	Accredited partner laboratory	Scope matched to requested tests, internal QC and external quality assurance
Mass spectrometry	Specialist laboratory; LC-MS/MS, GC-MS or ICP-MS	Internal standards, calibration, blanks and matrix specific validation
Cytometry	Specialist laboratory; selected immune or telomere questions	Panel and instrument controls, standardised analysis
Breath analysis	Collection room plus specialist lab	Validated bags or sorbents, timing, ambient blanks and method controls
Safety and waste	Collection site and lab; sharps, spill supplies and biomedical waste	Local clinical procedures and trained personnel
LIMS	Laboratory information management system and secure result gateway	Sample chain, audit history, result correction and authorised export

No clinic purchase of a full molecular laboratory is assumed. The initial equipment budget separates on site collection, processing actually performed on site and outsourced testing. High cost analysers enter a laboratory partner's capability register. Device and assay specifications, service arrangements and consumable costs precede any purchasing shortlist.

Milestones and comparability

A proposed schedule offers two baseline observations where practical, checkpoints around sessions 20 and 40, an end visit around session 60 and an agreed follow up. Each panel has its own schedule: genetics once; clinical markers where relevant; selected exploratory markers at baseline and end; acute saliva or breath studies only around specified sessions. This avoids treating every specimen as useful after every exposure.

Visits record date, time, time since session, diet or fasting where relevant, hydration, sleep, medication, recent illness, exercise and other chosen context. A 60 session programme is not necessarily 60 calendar days. Follow up uses actual elapsed time.

Matched specimen type, laboratory method and processing improve comparability. Samples from different timepoints are balanced across analytical batches where feasible, with controls and analysts blinded to condition where appropriate. The statistician defines detectable change and measurement error before calling a result meaningful.

A sample from invitation to result

The agreed test order specifies assay, specimen, kit, collection conditions and receiving laboratory. Staff confirm identity and current permission; collection generates a barcode and time record. Processing logs additive, volume, delays, centrifugation or stabilisation and aliquots. Transport records custody and temperature where required. Laboratory accession checks integrity and accepts, flags or rejects the specimen using its criteria.

The result retains value, unit, method, uncertainty or detection limit, laboratory flags and release status. Clinically relevant results follow an agreed clinician communication pathway. Exploratory outputs retain their research label. The participant sees a comprehensible explanation and chooses whether a selected result enters the Aura. Sample storage, deletion of digital copies and destruction of remaining physical material have separate records.

11 High value moments and model fidelity

Preserve enough for the chosen task

A high value moment might be a correction, a changed preference, an important memory, a prediction test or an unexpected response. Value is defined in relation to a task and the participant's priorities. Physiology never alone decides that a private memory deserves permanent retention.

The proposed capture system maintains a short rolling buffer only for consented streams, with its duration and overwriting behaviour disclosed. It preserves a bounded window around agreed events, plus scheduled neutral samples. Event selection logs why a window was retained. Neutral sampling provides a reference so the model does not learn only from dramatic moments.

Always collected at a chosen low burden level, selectively collected, and never collected are explicit per stream states. Essential clinical monitoring has its own record policy. A private moment marked “discard” stops optional retention and purges dependent features according to the disclosed retention plan.

Compression strategy

Original audio or observations, extracted features and generated summaries are separate layers. Source windows retain necessary calibration and context. Feature extraction records algorithm, version and parameters. Summaries link to sources and retain disagreements. A compact personal graph and retrieval index support everyday assistance; the archival store is retrieved only when relevant.

Model distillation or personal fine tuning is optional. Retrieval over a controlled record is the starting design because corrections are easier to trace and remove. Selected adapters or embeddings remain sensitive personal information. A compressed model includes a capability card describing what it covers, what was lost and where it abstains.

Evaluate distortion

Test	Comparison	Failure example
Meaning	Participant reviewed summary versus original	A wish to travel becomes a committed relocation plan
Memory	Retrieval answer versus source record	Invented events appear as autobiography
Prediction	Prospective forecast versus observed outcome	High confidence repeatedly exceeds actual accuracy
Personalisation	Selected setting versus later participant review	A disliked sound keeps returning
Uncertainty	Sparse evidence versus confidence	Unknown sexuality is silently filled in
Portability	Exported model versus source system	Permissions, corrections or timestamps disappear
Resource cost	Full archive versus compact model on the same task	Compression saves storage but makes the task unusable

Testing uses held out tasks and sessions, with participant review where meaning is involved. Self report has its own perspective and uncertainty; it is preserved as such. A disagreement creates parallel interpretations until resolved, rather than a forced single answer. No finite model captures every detail of a person.

12 Computing architecture and resource budgets

What compute means here

Compute is processing work performed by a phone, local computer, clinic server or larger shared service. Storage holds records; memory holds active work; a GPU accelerates selected model and rendering tasks. Networking connects them. More computing capacity does not by itself make a model faithful.

The system separates clinical control, sensor acquisition, game presentation, personal memory and optional research. A clinician's approved controller operates treatment. A read only bridge exports permitted observations to AuraOS. The game proposes suitable content within an approved library and participant choice; it has no command path to pressure, gas delivery or medical alarms.

Proposed component responsibilities

Component	Function	Boundary
Clinical controller	Pressure, gas, alarms and treatment record	Independent of game and general AI
Acquisition gateway	Timestamp and validate authorised sensor streams	No treatment commands; encrypted local buffer
Quest engine	Branches, scenes, input, accessibility and progress	Reads authorised context; logs content version

Component	Function	Boundary
Dialogue service	Speech, retrieval and participant conversation	Scoped memory access; separates proposals from facts
Personal vault	Original records, graph, corrections and permissions	Participant controlled policy and recoverable encryption
Model evaluator	Fidelity, calibration and compression tests	Versioned results; no hidden personality verdict
Lab bridge	Import authorised results and corrected reports	Coded identity mapping and verified laboratory provenance
Research workspace	Defined study dataset and reproducible analysis	Separate consent, access roles and retention
Federation service	Optional cross site analysis	Reviewed export, secure aggregation or privacy controls

Where the work runs

L0 is the person's private environment, including their chosen devices and protected personal context. A clinic may temporarily host L0 processing for a session, with a documented operator agreement. Physical proximity alone does not make a clinic computer personally sovereign.

At L1, an agreed local cooperative or clinic supplies shared capacity. L2 supports regional collaboration; L3 supports wider commons. These levels describe audiences and relationships, not automatic upload destinations. The “missing middle” is local shared compute between personal devices and hyperscale services. It supports larger models, creative work and research without forcing all private records into a distant service. [P2, P6, P7, P8]

A home device prioritises offline records, suitable compact models and sync chosen by the participant. A clinic workstation handles acquisition and presentation. A local server serves concurrent sessions through separated workspaces. Larger services receive only authorised jobs and specified inputs. Model availability, licence and data use terms are checked for each chosen provider.

Illustrative engineering profiles

Profile	Planning envelope	Intended work
Personal record and modest dialogue	16 to 32 GB RAM; encrypted storage; CPU or available accelerator	Retrieval, permissions, text and selected local inference
Clinic session workstation	32 to 64 GB RAM; 16 to 24 GB GPU memory where workload needs it; 1 to 2 TB fast scratch	Multimodal processing and a single rich session
Local shared server	128 to 256 GB RAM; accelerators and redundant storage sized after load tests	Several isolated sessions, lab import and research jobs
Archive and backup	Capacity based on retained media plus replication	Versioned vaults, recovery and agreed retention

These are proposed envelopes, not procurement recommendations or performance guarantees. A model with 8 billion parameters at 4 bit weight storage has roughly 4 GB of raw weights before runtime overhead, context

caches and other models. Its feasibility depends on software, device and concurrent workload. Unity or Unreal Engine are candidate presentation runtimes; the record schema stays independent of either engine. [P1, P7]

Storage worked example

Thirty two EEG channels at 500 samples per second with 4 bytes per stored sample produce 64,000 bytes per second: approximately 461 MB for a two hour session and 27.6 GB across 120 hours. One mono audio stream at 48,000 samples per second with 2 bytes per sample adds approximately 691 MB per session and 41.5 GB per programme before compression. One compressed video stream averaging 5 megabits per second adds 4.5 GB per session and 270 GB per programme. Decimal units are used.

Together these example streams total about 339 GB per participant before other sensors, metadata and backups. One hundred comparable participants total about 33.9 TB. Three complete retained copies require about 101.7 TB before headroom. Retaining only selected media windows changes this sharply; retention policy therefore precedes storage procurement. Sequencing data and multiple video views are separately budgeted.

Performance and energy

Proposed targets are a visible input response within 100 milliseconds, a dialogue acknowledgement within 500 milliseconds and a useful first response within about two seconds for ordinary local queries. Complex work gets a progress indicator and a choice to continue quietly. Actual latency is benchmarked with the selected model, speech pipeline and clinical room setup.

Acquisition remains stable if inference slows. Backpressure reduces optional media work before dropping essential event records; dropped frames and samples are explicit. Per session reporting includes energy, GPU time, storage growth and network transfers. An illustrative 300 watt average workstation load uses 0.6 kWh in two hours, or 36 kWh in 120 hours. This excludes chamber operation, cooling, laboratory work and shared servers. Heat reuse is a separate building engineering project.

13 Data sovereignty and the system of trust

Sovereignty in practical terms

A participant chooses what is observed, retained, interpreted, shared and used to train a model. Each choice names purpose, audience, duration and withdrawal effects. Health and genetic information have sensitive status under Australian privacy arrangements; the specific provider and jurisdiction determine applicable obligations. [E9]

A private dashboard answers: What is held? Where is it held? Who accessed it? Which conclusions depend on it? What changes if it is removed? Plain language controls accompany the underlying permissions. There is a viewable access history and an export route that does not depend on continuing the programme.

Separate permissions

Permission	Example	Independent choice
Live observation	Hear the microphone for this conversation	Does not imply retention
Record retention	Save this story in the personal vault	Does not imply research use

Permission	Example	Independent choice
Interpretation	Analyse these selected physiological windows	Does not imply an honesty judgement
Clinical record	Keep the treatment and relevant care record	Explained separately from optional game records
Truth Engine research	Include selected controlled exercises	Does not include everyday relationship monitoring
Biological collection	Take a specified sample for named tests	Does not include future unspecified assays
Biobank	Store remaining material for a defined period	Future use and destruction choices stated
Research dataset	Share a described subset with a named study	Separate from provider model training
Model training	Train a specified personal or shared model	Separate scope, version and withdrawal limits
Cultural sharing	Release material under custodial conditions	Individual agreement alone may be insufficient
Outside action	Let the Aura perform a named task	Scope and expiry separate from knowledge access

Identity protection and access

The personal vault uses encryption at rest and in transit, separated keys and role based access. Staff authenticate individually; researchers receive coded datasets with the identity mapping controlled separately. A lost key recovery plan balances participant control with access continuity. Trusted recovery contacts or escrow are chosen and documented; possession of a login alone does not confer access to every sensitive stream.

Clinic copies expire under the agreed plan after synchronisation and any required record retention. Backups have an expiry and recovery process. Support staff receive the minimum access needed for their task. Medical staff retain appropriate clinical access through the disclosed care pathway, without silently acquiring access to private creative or intimate records.

Sharing and federation

Federated learning sends model work to local data, but model updates may still reveal information. Aggregation, participant counts, attack testing and appropriate privacy methods are evaluated before sharing. “Anonymous” is never assumed from removal of a name. Small cohorts, rare stories, voice, face and genetics are especially identifying.

Peer to peer storage is optional and encrypted. Raw health records, biometrics, genetic data and sensitive metadata are not written to a public blockchain or openly addressable archive. A signature verifies origin and integrity, not truth. A hash may still support linkage to known content. Claims about decentralisation do not replace tested access control or deletion behaviour.

Corrections and withdrawal

Original observations, participant accounts and AI interpretations stay distinct. Corrections become new versioned entries and invalidate dependent summaries, embeddings, caches and queued exports. A corrected laboratory result replaces the displayed current result while preserving an appropriate clinical audit trail. Disagreement is visible until reviewed.

Withdrawal stops future optional collection and sharing. Unshared optional records and remaining samples follow the chosen deletion or destruction process. Clinical record retention, completed aggregate analyses, previously distributed copies and model training may limit reversal; these limits are explained before consent. Revoking a credential or publishing a tombstone does not retrieve all copies or erase knowledge already learned by another party.

Cultural and relationship privacy

A family story may reveal information about another person. The participant marks identities, exclusions and sharing boundaries. Restricted cultural knowledge follows custodial authority over representation, access and reuse. Genetic material also implicates relatives; return of findings and any family communication use an agreed clinical and cultural pathway. [P2, P7, E6]

14 Research design and evidence

Begin with a tractable feasibility study

The first study asks whether the experience is understandable, acceptable, faithful enough for selected tasks and technically reliable. It tests consent comprehension, correction uptake, sensor usability, collection burden, accessibility, offline continuity and adverse experience reporting. Sample size follows these questions and the precision needed, rather than an invented claim of statistical power.

A later clinical study defines intervention, comparator, population, primary outcomes and analysis plan before enrolment. Reflection alone, immersion, treatment and their combination are separable conditions when the question requires it. Changes over time in a single group do not identify which element caused them. Research uses the applicable NHMRC framework and local institutional review. [E2]

Keep adaptation interpretable

The game adapts topic, wording and pace within participant choice. Clinical exposures do not adapt through an unrestricted AI optimiser. Sensory research uses a bounded approved library; changes, order, duration and washout assumptions are logged. Any algorithm adapting experimental conditions has its own prespecified study design, limits and stopping rules.

Comparison and expectancy studies have separate agreements and debriefing where appropriate. The game never pretends to deliver a treatment that a participant is owed. Treatment settings and placebo conditions are handled only within an approved clinical research arrangement. A favourable response to an expected stimulus does not establish a general “mind over matter coefficient”.

Outcome families

Family	Example feasibility measure	Interpretation
Participation	Chosen sessions, burden and withdrawal reasons	Acceptability, not obedience
Relationship	Trust, comfort and usefulness reports	Participant experience over time
Model fidelity	Meaning preservation and corrected errors	Task specific performance
Engineering	Signal availability, drift and system uptime	Reliability of the actual setup
Biological change	Prespecified assays with repeat uncertainty	Clinical or exploratory result according to purpose
Function	Suitable wellbeing, cognition or daily life outcomes	Validity and practice effects considered
Truth Engine	Held out discrimination and calibration	Limited to tested conditions
Resource efficiency	Accuracy and usefulness per storage or energy cost	Compression tradeoff on defined tasks

Missing samples, inaccessible tests, low signal quality and skipped topics remain separate states. Analyses report multiplicity, batch effects and selection bias. Findings about one participant do not automatically become population conclusions. Subgroup claims need an adequate design and evidence.

Clinical boundary

This document is a game and system specification, not a treatment protocol. Pressure, oxygen fraction, exposure timing, UV, hypoxia, specimen volumes, alarm thresholds and clinical eligibility are defined by qualified teams and device specific procedures. A symptom or concerning result moves to the agreed care pathway. An AI explanation never silently becomes a diagnosis.

The product's intended purpose determines medical device assessment. A tool that diagnoses, predicts or guides treatment has a different regulatory pathway from ordinary reflection software, even if both share the same avatar. Model and feature updates are reviewed for changes in intended use. [E10]

15 Beyond Genesis and the larger quests

A portable personal companion

The Aura Passport is a proposed portable bundle of selected personal context, model versions, permissions, recovery information and session history. It is not a public identity score or a compulsory full health profile. A destination receives only what is agreed for its role. Participants review the bundle before transfer and test access on a second device before relying on it. [P1, P2]

The closing ceremony lets the participant choose a setting, music or story, review what changed, acknowledge what remains unknown and agree the Aura's continuing role. The twin is never declared complete at hour 120. The next invitation is a practical experiment in daily life.

The world beyond the mini game

Quest family	Example invitation	Information needed
Care and wellbeing	Support a chosen daily routine or meaningful activity	Relevant preferences; health data only where needed and agreed
Culture and memory	Visit a place, record a story or join an invited cultural activity	Custodial permissions and selected narrative
Learning and work	Try a skill or rehearse a workplace conversation	Interests, goals and chosen role
Music and media	Create a song, book scene or short film	Selected creative material and authorship permissions
Community	Join voluntary care, gardening or mentoring	A minimal participation record, not the private twin
Environment	Explore a local digital twin or citizen science activity	Suitable place and observation permissions
Civic reflection	Explore a public issue through a chosen scenario	A separate voluntary branch, never clinical eligibility

A C-hour records one verified hour of voluntary human contribution. It has no monetary equivalent, conversion, redeemability, asset status or wage substitution. It does not pay for Genesis, unlock treatment or stand in for employment. Communities choose recognition and reciprocal opportunities; actual funding is accounted for separately. Personal reflection is not automatically a verified community contribution. [P5]

GAJRA Earth supplies a broader invitation towards Joyful Responsible Abundance. The Web3 Sensorium and Sovereign Space Builder offer ways to connect private learning to places and shared tasks. These connections are optional. A clinical participant is not asked to endorse a political platform or join a wider organisation to receive care. [P2, P6, P7, P8]

16 Production plan and acceptance criteria

Build in reviewable stages

Stage	Deliverable	Review before progressing
Concept review	This GDD, participant walkthrough and defined research questions	Participant, cultural, clinical, laboratory and engineering input
Ordinary room prototype	Small quest library, private vault and simulated sensor replay	Choice, corrections, accessibility and traceable interpretation
Equipment bench work	Acquisition gateway with reference signals and external result import	Calibration, time alignment, failure handling and cybersecurity
Supervised feasibility	Defined participant study in an appropriate setting	Ethics and governance, burden, fidelity and usable signals

Stage	Deliverable	Review before progressing
Chamber integration	Approved read only bridge and permitted experience interfaces	Device specific engineering and clinical compatibility
Comparative research	Prespecified study with suitable control conditions	Analysis plan, outcomes, sample handling and independent review
Wider deployment	Repeatable site pack, training, support and portability	Demonstrated performance, approved intended uses and maintenance

No deployment dates, approved chamber status, partners, clinical outcomes or procurement prices are assumed. The plan advances through evidence and review of concrete outputs.

Functional acceptance tests

A participant pauses recording and sees optional capture stop. A skipped research component leaves the main quest map available. A corrected preference changes the next relevant suggestion. An invented scene retains its simulation label in export. A lost sensor produces a gap and quality flag rather than fabricated readings. A lab result with a changed identifier is quarantined for review rather than attached by a fuzzy name match.

A server outage leaves the clinical controller and alarms operating normally. A game process has no permission or physical interface to change treatment. Acquisition survives delayed inference and records loss explicitly. A permission revocation blocks future research exports and creates a dependency cleanup report. Export and reimport retain sources, units, corrections, consent and timestamps. A recovery drill restores an authorised vault without exposing another participant's records.

Content acceptance tests

The human's words remain distinguishable from an AI summary. A wish does not become a commitment. Symbolic lenses are labelled by purpose without belittling personal meaning. No questionnaire or physiological trace supplies an automatic moral, sexual or spiritual classification. The Truth Engine refrains from real world verdicts outside its tested scope. Biological dashboards preserve clinical and exploratory labels.

Delivery team and site pack

The proposed team includes game and accessibility design, software and security engineering, participant involvement, hyperbaric medicine where applicable, clinical research coordination, laboratory science, statistics and appropriate cultural custodians. Roles and actual appointments are established before a study.

A site pack includes equipment IDs and permitted configurations, cleaning and calibration schedules, acquisition settings, clinical interface permissions, quest versions, sample manuals, laboratory contacts, consent materials, incident routes, export procedures and restoration drills. Mass production targets repeatability and maintainability; it does not substitute for validation of each product configuration.

17 Technical records and reusable quest format

Core records

Record	Minimum content
Participant profile	Chosen identity, preferences, accessibility and separate linkage IDs

Record	Minimum content
Consent receipt	Scope, purpose, audience, date, expiry, version and withdrawal effects
Session	Start and end, setting, operator, game and equipment versions
Observation	Source, units, timestamps, quality, calibration and permissions
Moment	Event, original contribution, selected windows and participant interpretation
Claim	Author, evidence links, uncertainty, current status and revision history
Scenario	Assumptions, branch, simulated outputs and participant response
Quest	Purpose, content version, options, prerequisites and save policy
Sample	Coded identity, specimen, collection, processing, custody and consent
Assay result	Method, value, unit, laboratory, QC, release and correction history
Model version	Inputs, licence, training scope, evaluator results and resource budget
Export bundle	Selected records, recipient, purpose, expiry and portability manifest

The proposed schema supports ordinary structured exports, readable originals and established laboratory interfaces where available. Interoperability adapters map laboratory identifiers and units without discarding the original report. Sequence datasets retain suitable contextual metadata, informed by MIxS. [E11] Schema and vocabulary versions travel with the record.

A playable quest example

Quest G21 asks “What does time with family mean in practice?” Entry requires only participant interest. The scene is a chosen quiet space. The participant names a priority and an example. The Aura retrieves only authorised time records and offers a comparison with coverage visible.

Branch A accepts the comparison and explores a small action. Branch B adds a constraint or missing event. Branch C corrects the original priority. Branch D leaves the subject private. Branch E chooses quiet. Every branch offers a closing review; none awards more worth or greater access.

Optional sensors describe context only. The quest saves the person's account, the proposed comparison, any correction and a chosen next step. The success condition is that the participant understands and approves the retained interpretation, or declines retention. A follow up asks what happened without treating an unfinished plan as a personal failure.

A quest authoring template

Each quest includes a stable ID, title, purpose, source and licence, intended audience, estimated duration, accessibility variants, entry choice, prompt sequence, branches, exit options, optional streams, data retention, distress response, outside action permission and review criteria. Research quests also include protocol ID, experimental conditions, ground truth where available and debriefing. Clinical settings are referenced, never authored inside a quest.

18 Sources and evidence notes

Project documents

P1. AURA GEODE to MACRO, supplied October 2026 PDF, including earlier architecture and conversation material. Supplies the Geode, PADS, AuraOS, dialogue loop, 60 session arc, Bio-Kit and continuing relationship concepts. Proposed capabilities and historical wording are treated as design material.

P2. Joyful Responsible Abundance Human and SI Alignment Revised, supplied edition 1. Supplies the person and Aura relationship, CYOA, continuity, self alignment, meaning preservation and local compute context.

P3. Clinical Research Path for Aura of Dementia, supplied October 2026 PDF. Supplies clinical translation, feasibility, governance and dementia adaptation context; it does not establish validation of the Genesis system.

P4. 82 Claytons Road Anglican Partnership Proposal, edition 5. Supplies the proposed care setting, immersive reflection, memory palaces and participant centred Geode framing. It does not establish a confirmed site or clinical partnership.

P5. C Hour Joyful Responsible Abundance Plan, planning edition 5, 23 September 2026. Governs the definition of C-hours used here, including separation from financial accounting.

P6. Web3 Sensorium for Science Debate, supplied October 2026 PDF. Supplies private nodes, contextual sensing, shared scientific environments and provenance.

P7. Sovereign Space Builder part 2 Full System Specification, supplied October 2026 PDF. Supplies offline first nodes, L0 to L3 cooperation, typed records and custodial access concepts.

P8. UNGA81 Joyful Responsible Abundance Refined, edition 5. Supplies the broader alignment, cooperative infrastructure and missing middle compute context.

P9. Oceania Health and AI Surge Plan, supplied October 2026 PDF. Supplies ambitious deployment context. Its proposed rollout quantities, health effects and financial C-hour framing are not adoption assumptions in this GDD. P5 governs C-hour meaning; clinical effects remain study questions.

P10. Protopian Gambit Captions, supplied SRT. Supplies the surrounding narrative world and Genesis references; spoken assertions are not scientific validation.

External evidence checked on 13 October 2026

E1. Therapeutic Goods Administration. Risk of fire during use of hyperbaric chambers, safety advisory, 10 October 2025. Device instructions, training and environment compatibility guide chamber integration. https://www.tga.gov.au/safety/safety-monitoring-and-information/safety-alerts/risk-fire-during-use-hyperbaric-chambers

E2. NHMRC. National Statement on Ethical Conduct in Human Research 2025. The current statement and institutional procedures guide consent and research review. https://www.nhmrc.gov.au/about-us/publications/national-statement-ethical-conduct-human-research-2025

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The numeric sensing bands, timing goals, hardware envelopes and worked resource budgets in this document are proposed planning specifications. They await bench measurements, device selection and feasibility review. External references support the stated boundaries; they do not validate the proposed Aura Genesis system or its full combination of interventions.